

Package ‘cdmetapop.R’

November 21, 2024

Title Utility functions for CDMetaPOP outputs

Version 0.0.1-dev

Description This package provides a set of helper functions to analyze and visualize the output files of CDMetaPOP

Imports ggplot2,
reshape2,
tidyr

License friends to pick a
license

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

R topics documented:

age_plus_one_plot	2
age_structure_proportions	2
alleles_by_year	3
births	3
dispersal	4
extinct_patch	5
hets_plot	5
myy_progeny_plot	6
pop_count_plot	6
pop_mature_plot	7
pop_patch_count	7
pop_sex_plot	8
pop_year_class	8
read.cdmetapop	9
separate_column	9
size_age_class	10
unite_column	10
Index	11

age_plus_one_plot	<i>Function to extract population sizes of age +1 individuals</i>
-------------------	---

Description

This function produces a plot with population sizes of age 1+ individuals. It is based on the N_Initial_class column of the CDMetaPOP output file summary_classAllTime.

Usage

```
age_plus_one_plot(data)
```

Arguments

dataframe	summary_classAllTime.csv dataframe
-----------	------------------------------------

Value

ggplot

age_structure_proportions	<i>Determine the proportions of the age structure of the population</i>
---------------------------	---

Description

This function summarizes the proportions of the age structure in the simulated population. It is calculated counting the values in the column 'age' of the chosen 'ind' generation.

Usage

```
age_structure_proportions(path = mypath, runs = 1, gen = 49, species = 0)
```

Arguments

path	path to the simulation folder
runs	integer referring to the number of montecarlo simulations output files to examine. It refers to the mcruns option in the RunVars CDMetaPOP input file. Right now it defaults to 1, which would be only one montecarlo run.
gen	Simulation run time (generation or year). Refers to runtime in the RunVars of CDMetaPOP input file. Currently defaults to 49.
species	If the simulation includes more species... The code for this needs to be adjusted

Value

a dataframe with a column for each montecarlo run (MC) with the proportion of the population for each given age (in rows) and the last column with the average value across all montecarlo runs at any given age.

Examples

```
define your path to the output_test folder: mypath = "~/UM/CDMetaPop/EBT/EBT_Inputs/data/output_test172848770"
```

alleles_by_year	<i>Function extracts and plots allele count overtime</i>
-----------------	--

Description

This function plots the total number of unique alleles for every year of the simulation. It is based on the Allele column of the CDMetaPOP output file summary_popAllTime

Usage

```
alleles_by_year(data, n = 5)
```

Arguments

n	An integer that specifies the time intervals of the time series. It defaults to 5 years.
dataframe	summary_popAllTime.csv

Value

a ggplot object representing the overtime observed and expected heterozygosities values in the simulation.

births	<i>Create a plot with the proportion of young of the year. Note here that this proportion includes only age 0 progeny that survived egg mortality.</i>
--------	--

Description

This function plots the proportion of progeny over the total, after egg deaths are taken into account

Usage

```
births(file_path)
```

Arguments

x	Specify the path to the summary_popAllTime.csv dataframe
---	--

Value

a ggplot object representing the size of progeny surviving egg deaths every year

dispersal	<i>This function counts the number of individuals that move each year by reading in each ind file</i>
-----------	---

Description

This function counts the number of individuals that move each year by reading in each ind file

Usage

```
dispersal(
  path = path,
  run = 0,
  batch = 0,
  mc = 0,
  gen = 9,
  species = 0,
  ind = TRUE
)
```

Arguments

path	path to the simulation folder
run	A run is each row of the RunVars file - Fix description
mc	integer referring to the number of montecarlo simulations output files to examine. It refers to the mcruns option in the RunVars CDMetaPOP input file. Right now it defaults to 0, which would be only one run.
gen	Simulation run time (generation or year). Refers to runtime in the RunVars of CDMetaPOP input file. Currently defaults to 49
species	If the simulation includes more species... The code for this needs to be adjusted. Right now, species = 0, means one species only and it is the default
ind	If ind is true, the function will return calculations based on the ind files, otherwise on the sample files

Value

a dataframe The data returned is a dataframe that lists the proportion of dispersors for each year and for each mc simulation

Examples

define your path to the output_test folder: mypath = "your_path/your_output_test/" For example "~/UM/CDMetaPop

extinct_patch	<i>Check which patch(es) went extinct (no more individuals left) at a particular year</i>
---------------	---

Description

This function produces a vector with the number of patches that suffered extinction at a given year. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime.

Usage

```
extinct_patch(data, year = 100)
```

Arguments

year	value referring to the year to select. It defaults to 100
dataframe	summary_popAllTime.csv dataframe

Value

integer vector

hets_plot	<i>Create heterozygosity plot</i>
-----------	-----------------------------------

Description

This function plots heterozygosity overtime. It is based on the Ho and He columns of the CDMetaPOP output file summary_popAllTime

Usage

```
hets_plot(data)
```

Arguments

x	dataframe summary_popAllTime.csv
---	----------------------------------

Value

a ggplot object representing the overtime observed and expected heterozygosities values in the simulation.

myy_progeny_plot	<i>Plot the proportion of exclusive Myy progeny over the total</i>
------------------	--

Description

This function plots the ratio of the Myy progeny overtime

Usage

```
myy_progeny_plot(file_path)
```

Arguments

x Specify the path to the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime progeny size of YYMales

pop_count_plot	<i>Create abundance by year plot</i>
----------------	--------------------------------------

Description

This function plots population dynamics overtime showing a count of the individuals over the years of the simulation. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_count_plot(data)
```

Arguments

dataframe Import the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime general population size

pop_mature_plot	Create abundance by year plot showing the numbers of mature individuals of all sexes
-----------------	--

Description

This function plots population dynamics overtime showing a count of the mature individuals over the years of the simulation separated by their sex. It is based off the N_Mature columns of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_mature_plot(file_path)
```

Arguments

x	Specify the path to the summary_popAllTime.csv dataframe
---	--

Value

a ggplot object representing the overtime population size of sexually mature individuals

pop_patch_count	Create abundance by patch plot
-----------------	--------------------------------

Description

This function plots the population abundance per Patch. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime.

Usage

```
pop_patch_count(data, years = c(1, 25, 50, 100))
```

Arguments

years	vector of values referring to the years to select. It defaults to c(1,25,50,100)
dataframe	summary_popAllTime.csv dataframe

Value

a ggplot boxplot showing the population numbers of different patches

pop_sex_plot	<i>Create abundance by year plots for the different sex - Male/Female/MYY/FYY</i>
--------------	---

Description

This function plots population dynamics overtime showing the count of the individuals separated by sex over the years of the simulation. It is based off the following columns: N_Males_1, N_YYMales_1, N_Females_1 and N_YYFemales of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_sex_plot(data)
```

Arguments

dataframe	Import the summary_popAllTime.csv dataframe
-----------	---

Value

a ggplot object representing the overtime population size for the four sexes

pop_year_class	<i>Create abundance by year for each age class</i>
----------------	--

Description

This function plots population dynamics overtime according by the age of the individuals. It shows the count of the individuals over the years of the simulation separated by age. It is based off the N_Initial_Class column of the CDMetaPOP output file summary_classAllTime

Usage

```
pop_year_class(file_path, n = 5)
```

Arguments

n	An integer that specifies the time intervals of the time series. It defaults to 5 years.
x	Specify the path to the summary_classAllTime.csv dataframe

Value

a ggplot object representing the general population size at 5 years intervals for the different age classes.

read.cdmetapop	<i>Read cdmetapop file as dataframe</i>
----------------	---

Description

This function imports in r a cdmetapop output file as a dataframe

Usage

```
read.cdmetapop(file_path)
```

Arguments

x	Specify the file name and path.
---	---------------------------------

Value

Dataframe

separate_column	<i>Column symbol separator</i>
-----------------	--------------------------------

Description

This function eases the separation of column content generated from cdmetapop outputs. CD-MetaPOP uses delimiters and different delimiter options are specified for different fields. Delimiter 'l' or 'bar' is used time parameters. The delimiter '~' ('tilda') is used to assign parameters based on sex. The delimiter ';' is the default and used in various contexts to split fields and patches parameters. The delimiter ':' is used to split parameters used for functions. Note that Loo uses one instance of ';' in an Hindex option. See CDMetaPOP manual for more specific uses and examples.

Usage

```
separate_column(file_path, column_name, sep)
```

Arguments

x	Specify the file name and path.
---	---------------------------------

Value

Dataframe of columns for each value that was originally separated by the delimiter.

size_age_class	<i>Function to create a plot of size at age overtime</i>
----------------	--

Description

This function plots the overtime average size of individuals at a specific age. It is based on the AgeSize_mean columns of the CDMetaPOP output file summary_classAllTime

Usage

```
size_age_class(file_path, n = 5)
```

Arguments

x	Specify the path to the summary_popAllTime.csv dataframe
---	--

Value

a ggplot object representing the overtime average sizes of the individuals in the simulation separated by age.

unite_column	<i>Unite multiple columns into one</i>
--------------	--

Description

This replicates the function 'unite' from tidyr which can actually be used :)

Usage

```
unite_column(dataframe, column_name, sep)
```

Arguments

x	A data frame
---	--------------

Value

Description of the returned value.

Index

`age_plus_one_plot`, [2](#)
`age_structure_proportions`, [2](#)
`alleles_by_year`, [3](#)

`births`, [3](#)

`dispersal`, [4](#)

`extinct_patch`, [5](#)

`hets_plot`, [5](#)

`myy_progeny_plot`, [6](#)

`pop_count_plot`, [6](#)
`pop_mature_plot`, [7](#)
`pop_patch_count`, [7](#)
`pop_sex_plot`, [8](#)
`pop_year_class`, [8](#)

`read.cdmetapop`, [9](#)

`separate_column`, [9](#)
`size_age_class`, [10](#)

`unite_column`, [10](#)